

O-RAN White Box Hardware Working Group Outdoor Micro Cell Hardware Architecture and Requirements (FR1) Specification

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O-RAN White Box Hardware Working Group Outdoor Micro Cell Hardware Architecture and Requirements (FR1) Specification

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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

- x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).
- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the Outdoor Micro Cell deployment scenario case as specified in the Deployment Scenarios and Base Station Classes document [1].

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- [1] O-RAN-WG7.DSC.0-V01.00 Technical Specification, ‘Deployment Scenarios and Base Station Classes for White Box Hardware’.
- [2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [3] 3GPP TR 38.104: "NR; Base Station (BS) radio transmission and reception".
- [4] O-RAN-WG4.CUS.0-v01.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’.
- [5] CPRI Specification V7.0 (2015-10-09) Interface Specification, ‘Common Public Radio Interface (CPRI).
- [6] Small Cell Forum, SCF222.10.02 5G FAPI: PHY API Specification,
https://scf.io/en/documents/222_5G_FAPI_PHY_API_Specification.php
- [7] 3GPP TS 38.113: “NR: Base Station (BS) Electromagnetic Compatibility (EMC)
- [8] O-RAN Architecture Description, O-RAN-WG1-O-RAN Architecture Description - v01.00.00

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

Carrier Frequency: Center frequency of the cell.

F1 interface: The standard interface between an O-CU and an O-DU_x specified in [8].

Integrated architecture: An architecture wherein the O-RU_x and O-DU_x are implemented on one platform. Each O-RU_x and RF front end is associated with one O-DU_x and is connected to O-CU via F1 interface.

Split architecture: An architecture wherein O-RU_x and O-DU_x are physically separated from one another. A switch may aggregate multiple O-RU_x (s) to one O-DU_x. The O-DU_x, switch and O-RU_x (s) are connected by the fronthaul interface as defined in [4].

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

Occupied Bandwidth (OBW): It refers to the bandwidth occupied by the base station when operated, defined by the sum of the active bandwidths of the band allocation(s) operated. As defined by 3GPP TS 34.121 section 5.8, occupied bandwidth is the bandwidth containing 99% of the total integrated power of the transmitted spectrum, centered on the assigned channel frequency. The bandwidth between the 0.5% power frequency points is the occupied bandwidth.

Instantaneous Bandwidth (IBW): It refers to the bandwidth in which all frequency components can be simultaneously analyzed. It is defined by the frequency boundaries of the operating band(s).

Frequency Range: It refers to bandwidth defined by the frequency range within which the Base Station can be operated, defined by the band-pass filter of the BS; e.g., 3.4 – 3.8 GHz (400 MHz)

Frequency Band: A designated frequency range for the operation of the base station and the UE radios. 5G NR frequency bands are divided into two different frequency ranges: Frequency Range 1 (FR1) that mainly includes sub-6GHz frequency bands, some of which are bands traditionally used by previous standards but has been extended to cover potential new spectrum offerings from 410MHz to 7125MHz; Frequency Range 2 (FR2) that includes frequency bands from 24.25 GHz to 52.6 GHz. Bands in this mmWave range have shorter range but higher available bandwidth than bands in the FR1.

Fronthaul Gateway (FHGW): A fronthaul gateway is a physical entity that is located between a distributed unit and one or more radio units where it distributes, aggregates, and/or converts fronthaul protocols between the distributed unit and multiple radio units.

Normative terms used in this specification are defined in following:

Shall: indicates a requirement

Should: indicated as recommendation

May: indicates a permission

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
5GC	5G Core
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
ASIC	Application Specific Integrated Circuit
ATA	Advanced Technology Attachment
BPSK	Binary Phase Shift Keying
BS	Base Station
CDR	Clock/Data Recovery
CFR	Crest Factor Reduction
CU	Centralized Unit as defined by 3GPP
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DL	Downlink
DPD	Digital Pre-Distortion
DSP	Digital Signal Processor
DU	Distributed Unit as defined by 3GPP
DUC	Digital Up Conversion
EMC	ElectroMagnetic Compatibility
EVM	Error Vector Magnitude
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
FHGW _x	Fronthaul gateway with no FH protocol translation, supporting an O-DU _x with split option x and an O-RU _x with split option x, with currently available options 6→6, 7-2→7-2 and 8→8

1	FHGW _{x→y}	Fronthaul Gateway that can translate FH protocol from an O-DU _x with split option x to an O-
2		RU _y with split option y, with currently available option 7-2→8.
3	FPGA	Field Programmable Gate Array
4	GbE	Gigabit Ethernet
5	GNSS	Global Navigation Satellite System
6	GPP	General Purpose Processor
7	IEEE	Institute of Electrical and Electronics Engineers
8	IMD	InterModulation Distortion
9	I/O	Input/Output
10	JTAG	Joint Test Action Group
11	L1	Layer 1, also referred to as PHY or Physical Layer in OSI model
12	L2	Layer 2, also referred to as Data Link layer in OSI model
13	L3	Layer 3, also referred to as Network Layer in OSI model
14	LED	Light Emitting Diode
15	LTE	Long Term Evolution
16	MAC	Media Access Control
17	MCP	Multi-Chip Package
18	MIMO	Multiple Input Multiple Output
19	MTBF	Mean Time Between Failures
20	MU-MIMO	Multiple User MIMO
21	NG	Next Generation
22	NR	New Radio
23	OAM	Operations, Administrations and Maintenance
24	OBW	Occupied Bandwidth
25	O-CU	O-RAN Centralized Unit as defined by O-RAN
26	O-DU _x	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as
27		defined by WG4) or 8
28	O-RU _x	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined
29		by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same
30		at the O-DU _x
31	PCIe	Peripheral Component Interface Express
32	PDCP	Packet Data Convergence Protocol
33	PHY	Physical Layer, also referred as L1
34	PLL	Phase Locked Loop
35	POE	Power over Ethernet
36	QAM	Quadrature Amplitude Modulation
37	QPSK	Quadrature Phase Shift Keying
38	RAN	Radio Access Network
39	RF	Radio Frequency

1	RLC	Radio Link Controller
2	RRC	Radio Resource Controller
3	RU	Radio Unit as defined by 3GPP
4	RX	Receiver
5	SDU	Service Data Unit
6	SFP	Small Form-factor Pluggable
7	SFP+	Small Form-factor Pluggable Transceiver
8	SoC	System on Chip
9	SPI	Serial Peripheral Interface
10	TR	Technical Report
11	TRP	Total Radiated Power
12	TS	Technical Specification
13	TX	Transmitter
14	UL	Uplink
15	USB	Universal Serial Bus
16	WG	Working Group

Chapter 2 Deployment Scenarios and White Box Base Station Architecture

This chapter consists of two parts: the deployment scenario and the white box architecture. The deployment scenarios outline more specific functional requirements of the base station. All the reference designs shall meet these requirements in order to comply with O-RAN white box standard. In the white box hardware architecture section, it describes the overall gNB hardware architecture and function partition that meet the design requirements. The details on each of these topics are described in the following sections.

2.1 Deployment Scenarios

The specification here addresses a specific deployment scenario defined by the white box Deployment Scenarios and Base Station Classes document [1]. Its requirements are also defined in [1]. Some of the key requirements described in that document are highlighted below:

- Cell type: Outdoor Micro Cell
- Mounting Options: Rooftop, side of building/wall, pole
- Inter site distance: 1.5 kilometers
- Possible Carrier Frequency Bands: n2, n4, n5, n13, n41, n48, n66, n77, n78, n79
- [Instantaneous] Bandwidth: Up to 280 MHz
- Antennas Configuration: Up to 16T16R
- Coverage: Sectorized
- Conducted Power: up to 10W per port
- Number of Layers: Up to 16 layers
- Fronthaul Type: O-RAN FH (WG4) split option 7.2x.
- Architecture: Split architecture

2.2 White Box Base Station Architecture

In general, the base station hardware architecture can be classified by using different criteria. The physical partition method is adopted by O-RAN; hence, the base station architecture is divided into two categories namely split and integrated. In split architecture, the fronthaul interface determines the gNB functions location. Here we refer to the two partitions as O-DU_x and O-RU_x; where the “x” is split option number. In case the O-RAN WG4 defined fronthaul interface is used, these two partitions are called O-DU₇₋₂ and O-RU₇₋₂. For a more complete description of the terminology used, refer to the Deployment Scenarios and Base Station Classes Document [1].

The split architecture is shown in Figure 2-1. A Fronthaul Gateway (FHGW) is an optional device between the O-DU_x and O-RU_x to aggregate multiple radio units together as shown in Figure 2-5. Within this specification, a Back End fronthaul interface is defined as the connection between the FHGW_x and the O-DU_x, while the Front

End fronthaul interface is defined as the connection between the FHGW_x and the O-RU_x. With split architectures, one may choose to have O-CU and O-DU_x as either co-located or shared configuration with respect to hardware.

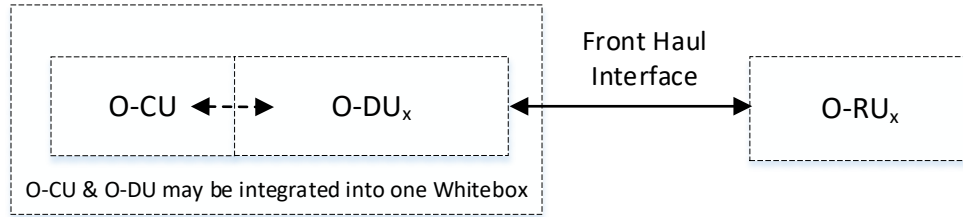


Figure 2-1: Split Architecture

For integrated base station architecture, the O-DU is integrated with the O-RU into one box. Figure 2-2 shows the integrated architecture.

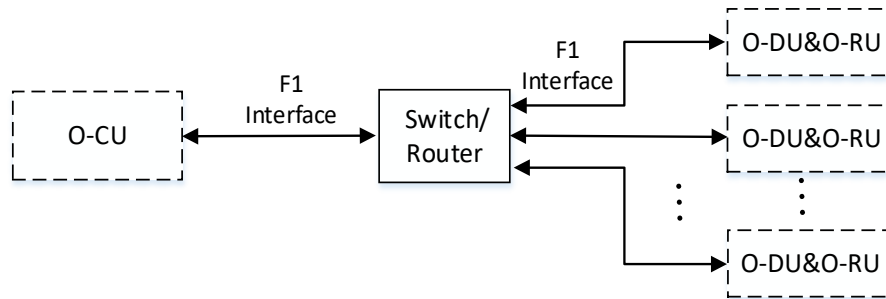


Figure 2-2: Integrated Architecture

2.2.1 Split RAN Architecture

For medium and large coverage deployment scenarios, the base station with split RAN architecture is more cost effective. This type of architecture is widely deployed in 4G networks. The main idea of this architecture is “shared cell”. By using the fronthaul gateway, many radio units can then share the same radio spectrum resource in one cell. This configuration is very useful in both low and high capacity scenarios. Therefore, when an O-DU_x and FHGW_x are capable of handling cell splits, multiple O-RU_x(s) can be grouped together to share the radio resources. Accordingly, the choice of fronthaul split option changes the architecture of the base station. WG7 has recognized the following four split option architectures per service providers’ deployment needs which are shown in Figure 2-3 through Figure 2-6.

1) Option 6 to option 6 split architecture

In split option 6, all L1 functions are within O-RU₆ while the O-DU₆ contains functions of MAC and above. Figure 2-3 shows the block diagram of base station having an option 6 split architecture.

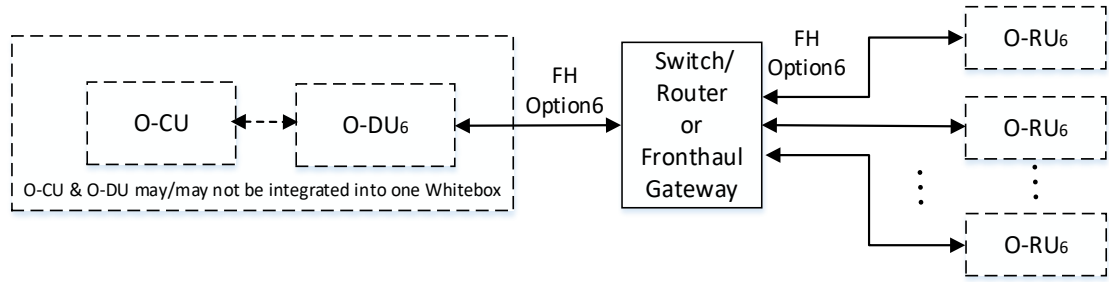


Figure 2-3: Option 6 to Option 6 Split Architecture

2) Option 7-2 to option 7-2 split architecture:

In split option 7-2, low PHY functions reside in the O-RU₇₋₂, while the high PHY functions reside in the O-DU₇₋₂. O-RAN WG4 CUS-plane spec [4] outlines the details of this split option under different usage scenarios.

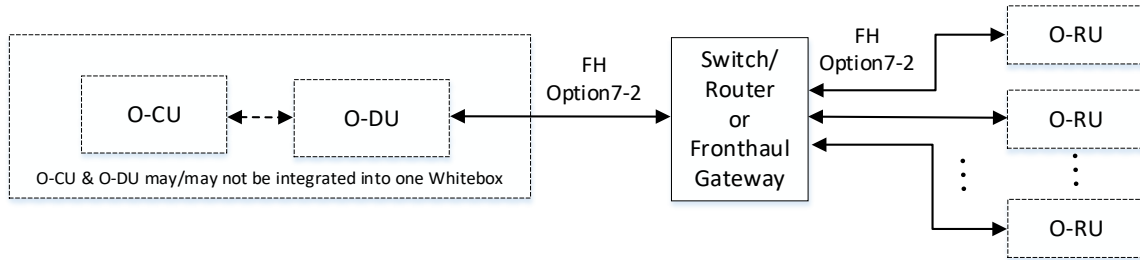


Figure 2-4: Option 7-2 to Option 7-2 Split Architecture

3) Option 7-2 to option 8 split:

This configuration is selected when a deployment scenario requires a radio unit which only supports split option 8 architecture. Since this is currently not part of overall O-RAN architecture, CPRI based option 8 fronthaul and FHGW_{7-2→8} will be included in the reference design specification. Figure 2-5 depicts the option 7-2 to option 8 based split architecture. There is no change for the definition of O-DU₇₋₂. However, O-RU₈ supports option 8 fronthaul interface while the FHGW_{7-2→8} translates the fronthaul protocol between option 7-2 and CPRI.

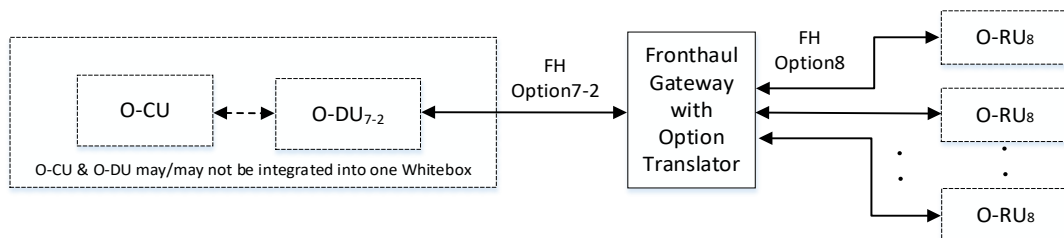


Figure 2-5: Option 7-2 to Option 8 Split Architecture

4) Option 8 to option 8 split architecture:

Currently, Option 8 is a non-O-RAN defined split option where the CPRI fronthaul interface is needed in order to make the interoperability work with O-DU₈ and O-RU₈ from different vendors. The option 8 fronthaul interface definition and the requirements shall be part of the white box reference design, if adopted. In this case, the O-DU₈ consists of L1 and L2 processing functions. Figure 2-6 shows an option 8 based split architecture. Note that O-CU and O-DU₈ hardware may be integrated into one Whitebox this is also shown in Figure 2-6.

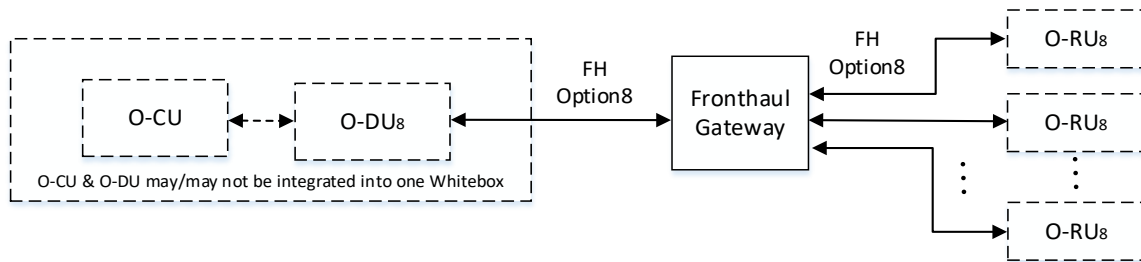


Figure 2-6: Option 8 to Option 8 Split Architecture

2.2.2 Integrated gNB-DU Architecture

For integrated gNB-DU, the complete L1/L2 and radio functions are integrated into a single box which includes all logical functions of O-DU and O-RU. Figure 2-7 shows the block diagram of integrated gNB-DU. The gNB-DU connects with the O-CU through an F1 interface as defined by [8]. This specification does not support an integrated option [per requirements in Section 2.1].

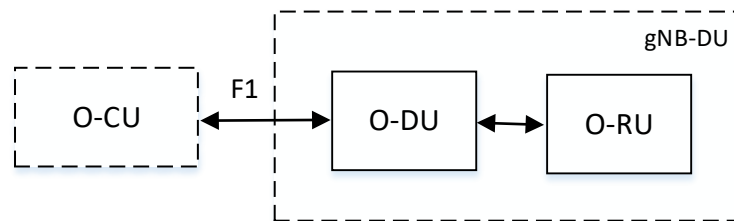


Figure 2-7: Integrated gNB-DU Architecture

Chapter 3 White Box Hardware Architecture

Based on the gNB physical implementation architectures discussed earlier, this chapter provides the architecture, major building blocks and all external/internal interfaces for each Whitebox.

3.1 O-CU Hardware Architecture

In a 3GPP system architecture, the gNB Central Unit (CU) communicates to the Distribution Unit (DU) via an F1 interface. This interface has been adopted by O-RAN Alliance as well. F1 is an IP based protocol interface, which offers more flexibility in the O-CU platform design.

3.1.1 O-CU Architecture Diagram

The O-CU can be implemented with any General-Purpose Processor (GPP) based platform having an optional hardware accelerator block. The O-CU functions can be implemented either on separated hardware platforms or on the same hardware platforms integrated with O-DU_x functions. In either case, the O-CU hosts L2/L3 functions, whereas O-DU hosts L2/L1 functions which require different CPU, Storage, Acceleration and Networking capabilities. Figure 3-1 shows the hardware blocks and interfaces within the O-CU white box.

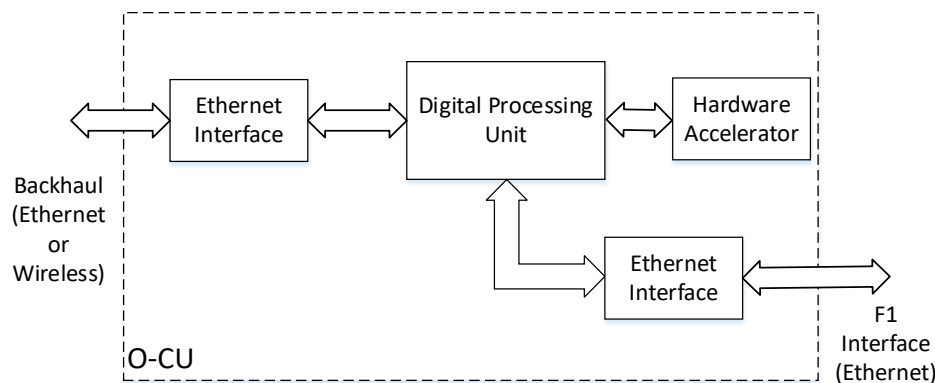


Figure 3-1: O-CU White Box Hardware Block Diagram

3.1.2 O-CU Functional Module Description

The O-CU functional architecture comprises Digital Signal Processing, optional HW Acceleration, and Connectivity (GbE) units as well as memory and storage units as shown in Figure 3-1.

3.1.3 O-CU Interfaces

The O-CU interfaces through backhaul with the vEPC or 5GC core network and via F1 interface with the O-DU₇₋₂. The backhaul and the F1 interfaces are typically implemented with GbE transport/connections.

3.2 O-DU₇₋₂ Hardware Architecture

For a split RAN architecture, the functional blocks of the RAN physical layer are divided into two parts – high PHY and low PHY. The O-RAN lower layer split is defined in O-RAN WG4 fronthaul interface specification [4]. It also provides the details of the interface protocol as well as function partitions.

3.2.1 O-DU₇₋₂ Architecture Diagram

Depending on the 3GPP standards and category of the radio unit, the split function blocks within O-DU₇₋₂ and O-RU₇₋₂ may vary accordingly. The O-RAN fronthaul C/U/S-plane specification [4] provides comprehensive information on this topic. The hardware functional partition architecture is shown in Figure 3-2. The High-PHY functions in O-DU are run on Digital Processing Unit, Hardware Accelerator, or both.

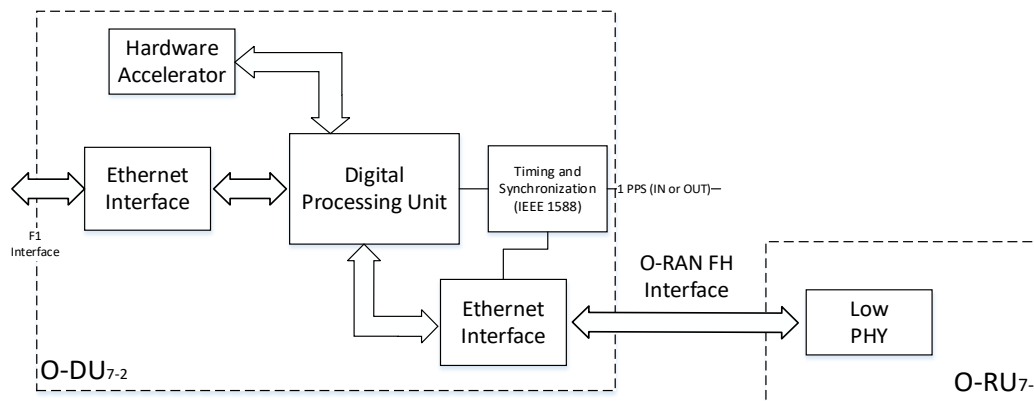


Figure 3-2: O-DU₇₋₂ with Split Physical Layer Function

3.2.2 O-DU₇₋₂ Functional Module Description

The choice of O-DU₇₋₂ hardware components depends on the product design requirements and is outside the scope of this specification. Figure 3-3 shows the main components of O-DU₇₋₂. Their descriptions and requirements are as follows:

Digital Processing Unit

1 The processing unit can be any GPP, FPGA or digital signal processor (DSP), with I/O chipset. It may also be in
2 the form of System-On-Chip (SOC), or Multi-Chip Package (MCP).

3 **Memory**

4 DDR memory devices are used to store the runtime data and software for the processing unit.

5 **Flash Memory**

6 On board non-volatile storage device is used to store the firmware and non-volatile data, such as log data.

7 **Board Management Controller**

8 The controller is used to manage/control the power and monitors the operational status of the board.

9 **Storage Device**

10 The storage device such as hard drive is used to store OS, driver and applications software.

11 **Ethernet Controller**

12 The Ethernet ports transport the fronthaul or backhaul traffic according to the gNB hardware node requirements.
13 The Ethernet device shall support IEEE1588 based timing synchronization.

14 **Hardware Accelerator**

15 The hardware accelerator is an optional device. For performance improvement, the hardware accelerator can be
16 used to process computationally-intensive functions (e.g., Forward Error Correction (FEC)) and to offload the
17 processor.

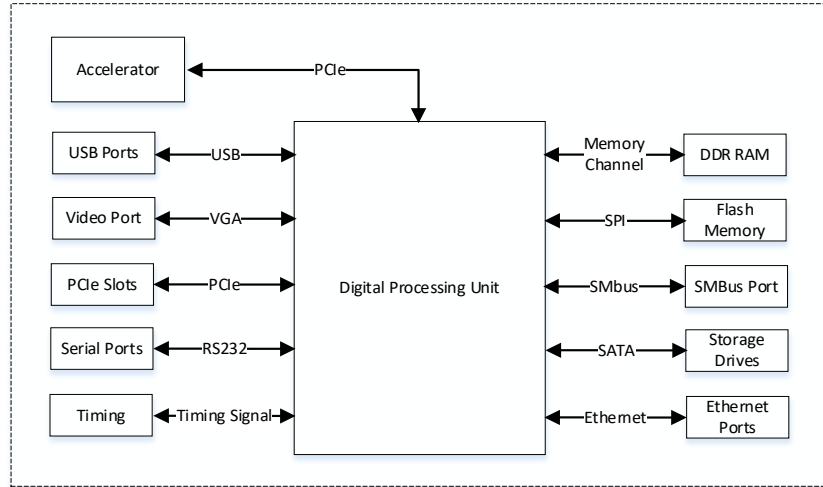


Figure 3-3: O-DU₇₋₂ Functional Block and Interface Diagram.

3.2.3 O-DU₇₋₂ Interfaces

The O-DU₇₋₂ supported interfaces described below are also shown in Figure 3-3.

Memory Channel Interface

Support DDR4 and later memory interface.

PCIe Interface

Support for PCIe Gen3 x16 and later interface versions; the bandwidth depends on the use cases, and it can be used to connect an accelerator device or network card.

Ethernet Interfaces

Supports any one or combination of GbE/10GbE/25GbE/40GbE links.

Serial ATA Interface

SATA3 shall be supported in case of software storage, such as hard drive.

SPI Interface

The SPI interface connects the processor with flash type of device for firmware.

Video Interface

Video interface is optional

USB Interface

Used to connect with local device for debug or on-board firmware update.

Miscellaneous Interface

Other interfaces that may be needed such as serial port, JTAG, etc.

3.3 O-DU₆ Hardware Architecture

This section intentionally left blank.

3.4 O-DU₈ Hardware Architecture

This section intentionally left blank.

3.5 FHGW₇₋₂ - Option 7-2 to Option 7-2 Hardware Architecture

This section intentionally left blank.

3.6 FHGW₆ - Option 6 to Option 6 Hardware Architecture

This section intentionally left blank.

3.7 FHGW_{7-2->8} - Option 7-2 to Option 8 Hardware Architecture

This section intentionally left blank.

3.8 FHGW₈ - Option 8 to Option 8 Hardware Architecture

This section intentionally left blank.

3.9 O-RU₇₋₂ Hardware Architecture

O-RU₇₋₂ consists of four modules, namely optical interface, digital processing unit, RF processing unit and antenna unit, as well as a timing unit as shown in Figure 3-4. The Ethernet interface complies with O-RAN WG4 open fronthaul interface. The O-RU₇₋₂ can directly connect with an O-DU₇₋₂ or connect through a FHGW₇₋₂. While the O-RU₇₋₂ hardware supports NR by default, LTE support is not precluded.

3.9.1 O-RU₇₋₂ Architecture Diagram

The O-RU₇₋₂ hardware architecture consists of the digital processing unit which handles all digital signal and interface processing and an RF processing unit handling all analog. There is a data conversion block for each transceiver path after the digital processing block which converts between digital signals and analog signals. A frequency conversion or mixer unit may be used to convert the IF or baseband frequencies to RF frequencies and vice versa. In some cases, a direct RF sampling transceiver is used, where no external mixer is required. The PA/LNA amplifies the RF signal, and the antenna module is used to transmit and receive signal over the air. There is at least one Ethernet port available which is used as the O-RAN fronthaul interface.

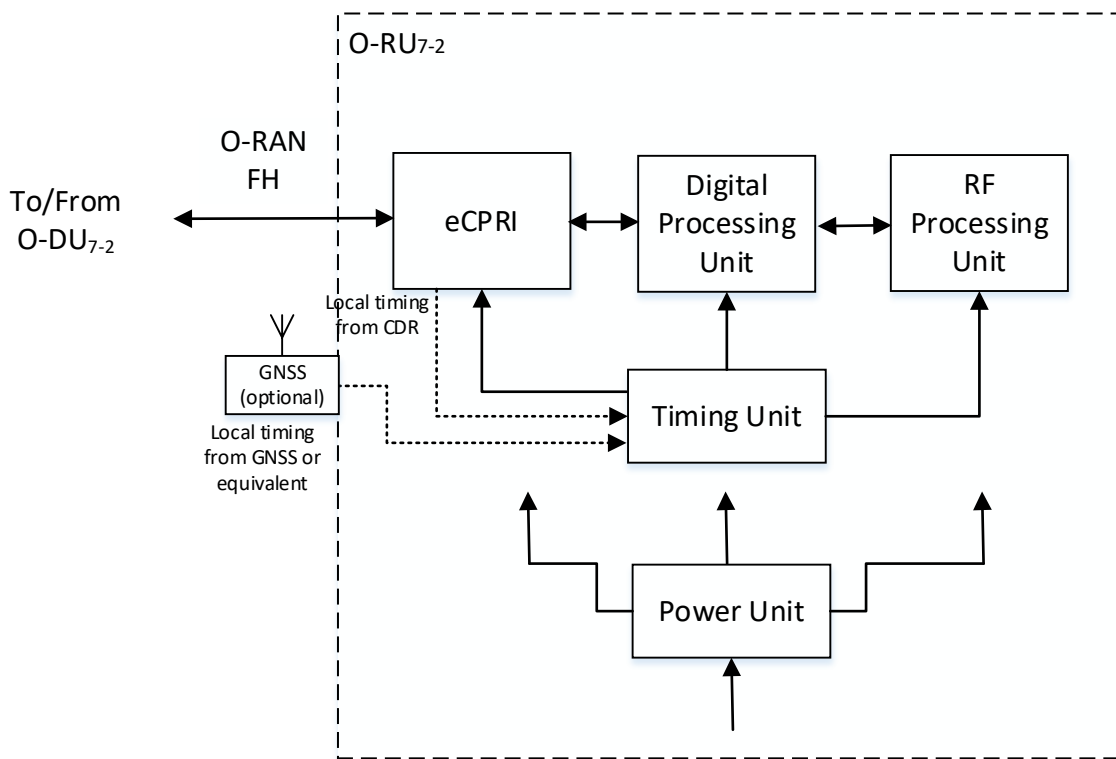


Figure 3-4: O-RU₇₋₂ Architecture Diagram

3.9.2 O-RU₇₋₂ Functional Module Description

In Figure 3-5, we illustrate O-RU₇₋₂ functional blocks that support O-RAN FH with split option 7-2x. There is at least one interface port which supports all fronthaul interface and PoE functionalities. The digital processing unit block of O-RU₇₋₂ is responsible for low-PHY functions such as FFT/IFFT, CP addition/removal, and PRACH

filtering. Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD) functions are also performed in digital processing unit. For bandwidth reduction, O-RU₇₋₂ architecture also supports the optional compression and decompression functions of FH interface. The ADC and DAC are mixed signal devices responsible for conversion of data between the digital and analog domains. As such, this block can be included as part of the either the digital processing unit or the radio processing unit. The RF Processing Unit consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low Noise Amplifier (LNA) and TX/RX filters. An antenna module follows that comprises physical antennae, RF feed distribution/aggregation network, and calibration network. The timing unit includes any clock and frequency synthesis required as well as other timing and synchronization circuits. In any given implementation, the physical boundaries between blocks may be slightly different, but the logical interfaces between will generally follow those shown in Figure 3-5.

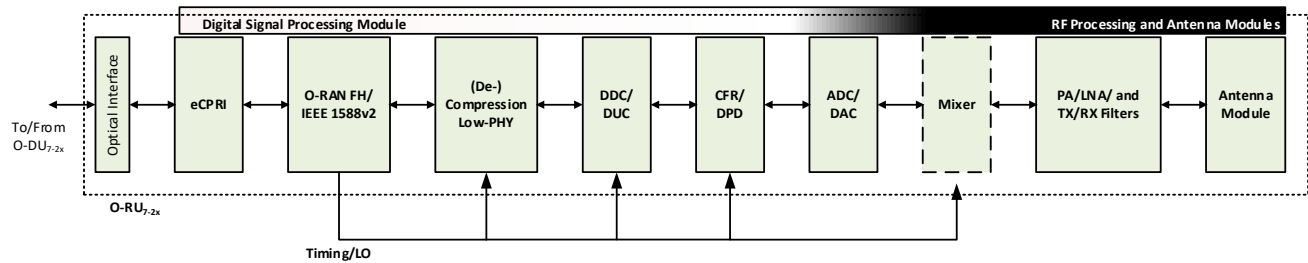


Figure 3-5: O-RU₇₋₂ Generic Functional Module Diagram

3.10 O-RU₆ Hardware Architecture

This section intentionally left blank.

3.11 O-RU₈ Hardware Architecture

This section intentionally left blank.

3.12 Integrated gNB-DU Hardware Architecture

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Chapter 4 White Box Hardware Requirements

This chapter provides the requirements for various white boxes used within the Outdoor Microcell base station. These white boxes are O-CU, O-DU_x, O-RU_x and FHGW_x. The O-CU and O-DU_x can be implemented in an integrated fashion into one white box hardware or they can be separated. The reference design based on specification shall meet all these requirements based on their priority as specified by operators.

4.1 O-CU Requirements

O-CU requirements are described in the following sections, which include the performance, interface, environmental, EMC, mechanical, thermal and power requirements.

4.1.1 O-CU Performance

The performance requirements of the O-CU are listed in Table 4-1.

Table 4-1 :O-CU Performance Requirements

Parameter	Requirement	Description	Priority
Synchronization	Support 1588v2 Synchronization	Timing synchronization method	High
	Support GNSS Synchronization		High
	Support BeiDou Synchronization		Low
	Support Galileo Synchronization		Low
	Support GLONASS Synchronization		Low
	Support switching between 1588v2 and GNSS		High
	Support SyncE		Low
Number of O-DU _x supported	For O-CU/O-DU _x integrated architecture: N/A	Number of O-DU _x (s) connected to a single O-CU	None
	For O-CU/O-DU _x non-integrated architecture: Determined by operator deployment requirements		High
Supported Cell Number	For O-CU/O-DU _x integrated architecture: N/A	Cells supported by an O-CU	None
	For O-CU/O-DU _x non-integrated architecture: Determined by operator deployment requirements		High

Supported RRC link number	For O-CU/O-DU _x integrated architecture: N/A	RLC links supported by O-CU	None
	For O-CU/O-DU _x non-integrated architecture: Determined by operator deployment requirements		High
Throughput between O-DU _x	For O-CU/O-DU _x integrated architecture: N/A	Throughput between O-CU and O-DU _x link	None
	For O-CU/O-DU _x non-integrated architecture: Determined by operator deployment requirements		High
Peak Data Rate	TBD	Highest theoretical data rate in error free conditions.	None
Latency	Control Plane < 10ms (def: Msg1 to Msg5) User Plane DL < 4ms UL < 4ms (def: PDCP SDU - > PDCP SDU)	O-CU latency	High

4.1.2 O-CU Interfaces

The interface requirements of the O-CU are listed in Table 4-2: O-CU Interface Requirements.

Table 4-2: O-CU Interface Requirements

Parameter	Requirement	Description	Priority
Transport Interface	10 GbE F1 Interface to connect with O-DU _x or 25 GbE F1 Interface to connect with O-DU _x	O-CU transport links to O-DU _x and 5GC	High
	10 GbE NG interface to connect with 5G core ^(b) or 25 GbE NG interface to connect with 5G core		High

4.1.3 O-CU Environmental and EMC

The EMC requirements of the O-CU are listed in Table 4-3.

Table 4-3: O-CU EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Shall comply with the requirements of 3GPP TS 38.113 (2017-12 Rel-15) [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

4.1.4 O-CU Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-CU are listed in the following tables.

Table 4-4: O-CU Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimensions	Built in any 19" standard rack, or stand alone, with overall height of no more than 5U ^(a) and depth (including the connector) less than 750mm.	Measurement in three dimensions	Low
Status Indicator LED	Shall include as a minimum the following status indicators: ---1 indicating on/off status of the power supply ---1 indicating on/off status of the transmission link	Indicator light	High

Note:

(a) Note that this dimension is application dependent and its value may change accordingly.

Table 4-5: O-CU Thermal Requirements

Parameter	Requirement	Description	Priority
Temperature and Humidity	Operate steadily and reliably over a long period of time under the following environmental conditions for data center: Operating Temperature: - 0 °C ~ + 50 °C Relative humidity: 15% ~ 85%	Environmental requirements for reliability	High
	The solution should support extended temperature range (-40°C to +65 °C) and humidity (5%-95%) if implemented outdoors.	Environmental requirements for reliability	Low

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

Table 4-6: O-CU Power Requirements

Parameter	Requirement	Description	Priority
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Power Supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300v, frequency range of 45Hz ~ 65Hz.	Power supply for O-CU	High
Power Dissipation	< TBD W full load operation	O-CU Power Requirement	High

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

4.2 O-DU_x Common Requirements

The O-DU_x here takes the function of high PHY or whole PHY, OAM function and layer 2. Usually the hardware is placed in the machine room, which can be collocated in the coverage building or in the central machine room far away from the coverage building.

4.2.1 O-DU_x Performance

The performance requirements of the O-DU_x are listed in Table 4-7.

Table 4-7: O-DU_x Performance Requirements

Parameter	Requirement	Description	Priority
Channel Bandwidth	≤200 MHz OBW	RF Bandwidth	High
Antenna Configuration (Number of Transceivers)	4T4R	Tx/Rx antenna number	High
	up to 16T16R		Low
Transmission distance	Measured in delay. TBD	Distance between DU and FHGW _x /RU	High
Connect Topology	NA	Fronthaul Gateway connection topology	None
Synchronization	Support 1588v2 Synchronization	Timing synchronization method	High
	Support GNSS Synchronization; Sync. between BS ≤ 150 ns; Carrier freq. error within one subframe <±0.1 PPM		High

	Support BeiDou Synchronization		Low
	Support Galileo Synchronization		Low
	Support GLONASS Synchronization		Low
	Support switching between 1588v2 and GNSS		High
	Support SyncE		Low
Capacity/MIMO	4T4R: 4 layers and 200 MHz OBW	MIMO related capability	High
	up to 16T16R and TBD		Low
Peak Rate	TBD	Peak data rate	High
Modulation	DL: QPSK,16QAM,64QAM,256QAM UL: $\pi/2$ -bpsk, QPSK, 16QAM, 64QAM	Modulation schemes	High
	UL: 256QAM		Medium
Latency	Control Plane< 10ms (def: Msg1 to Msg5) User Plane DL < 4ms UL< 4ms (def: PDCP SDU-> PDCP SDU)	Control/user plane Latency	High

4.2.2 O-DU_x Interfaces

The interface requirements of the O-DU_x are listed in Table 4-8.

Table 4-8: O-DU_x Interface Requirements

Parameter	Requirement	Description	Priority
Transport Interface	Fronthaul Gateway: N/A	O-DU _x transport links	None
	At least one 10 GbE F1 interface to connect with O-CU ^(b)		High

Notes:

(a) For O-CU and O-DU_x that are co-located

(b) For O-CU and O-DU_x that are separated physically

4.2.3 O-DU_x Environmental and EMC

The EMC requirements of the O-DU_x are listed in Table 4-9.

Table 4-9: O-DU_x EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Complying with the requirements of 3GPP TS 38.113 (2017-12R15) [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High

4.2.4 O-DU_x Mechanical, Thermal and Power

The mechanical requirements of the O-DU_x are listed in Table 4-10.

Table 4-10: O-DU_x Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimension	Built in any 19" standard rack, or stand alone, and the height does not exceed 5U ^(a) , the depth (including the connector) must be less than 750mm.	Measurement in three dimensions	Low
Status Indicator LED	Shall include as a minimum the following status indicators: ---1 indicating the on/off status of optical fiber interface ---1 indicating off status of the power supply ---1 indicating on/off status of the transmission link	Indicator lights	High

Note:

(a) Note that this dimension is application dependent and its value may change accordingly.

The thermal requirements of the O-DU_x are listed in Table 4-11.

Table 4-11: O-DU_x Thermal Requirements

Parameter	Requirement	Description	Priority
Temperature and Humidity	The O-DU _x shall operate steadily and reliably over a long period of time under the following environmental conditions for data center: Operating Temperature: - 0 °C ~ + 50 °C Relative humidity: 15% ~ 85%	Environmental requirements for reliability	High
	The solution should support extended temperature range (-40°C to +65 °C) and humidity (5%-95%) if implemented outdoors.	Environmental requirements for reliability	Low

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2 The power requirements of the O-DU_x are listed in Table 4-12.

3

Table 4-12: O-DU_x Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300v, frequency range of 45Hz ~ 65Hz.	Power supply for O-DU _x	High
Power Dissipation	< TBD W, with 4 cells full load operation	O-DU _x Power Requirement	High
	< TBD W, with 8 cells full load operation		Medium

4

5 4.3 O-DU_x Split Option Specific Requirements

6 In addition to the common O-DU_x requirements, there may be some specific requirements that apply to O-DU_x
7 due to the split option. These requirements are listed in the following sections.

8 4.3.1 O-DU₇₋₂ Specific Requirements

9 The specific requirements of O-DU₇₋₂ are listed in Table 4-13.

Table 4-13: O-DU₇₋₂ Specific Requirements

Parameter	Requirement	Description	Priority
Transport Interface	At least 4 Ethernet interfaces connected with Fronthaul Gateway	O-DU ₇₋₂ fronthaul transport links	High

4.3.2 O-DU₆ Specific Requirements

This section intentionally left blank.

4.3.3 O-DU₈ Specific Requirements

This section intentionally left blank.

4.4 O-RU_x Common Requirements

The O-RU_x common requirements apply to all the radio units regardless of the split option. In the indoor environment, O-RU_x hardware is placed on the cell or wall of the coverage building; it converts the base band signal into RF signal or vice versa to supply the coverage.

4.4.1 O-RU_x Performance

The O-RU_x performance requirements cover all the aspects of radio unit including frequency bands, antenna configurations, power efficiency, etc. Table 4-14 lists the performance parameters related to O-RU_x.

Table 4-14: O-RU_x Performance Requirements

Parameter	Requirement	Description	Priority
Operating band	n77	Radio frequency band	High
	n2, n4, n13, n41, n48, n66, n78, n79		Low
[Occupied] Bandwidth	≤200 MHz	Frequency bandwidth	High
Antenna Configuration (Number of Transceivers)	4T4R	Tx/Rx Antenna numbers	High
	up to 16T16R		Low

Output Power Accuracy	Under normal condition: $\pm 2\text{dB}$	Power accuracy	High
Tx off Power Level	less than -85dBm/MHz ; See section 6.4.1 in 3GPP TS 38.104 [3].	Tx power level during an off period	High
EVM at maximum output power	64QAM: EVM smaller than 8% 256QAM: EVM smaller than 3.5%	Max output power	High
Operating band unwanted emissions	The Operating band unwanted emissions shall satisfy the Category B limit defined by the section 6.6.4.2.4 in 3GPP TS 38.104 [3].	RF operation band unwanted emissions requirement	High
Transmitter spurious emissions	The Operating band unwanted emissions shall satisfy the Category B limit defined by the section 6.6.4.2.4 in [3].	Deliberately generated RF signal by transmitter	High
Receiver Sensitivity	The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel of G-FR1-A1-5; the reference sensitivity levels shall be better than -95.6dBm .	The weakest signal the receiver can identify and process	High
Blocking	In Channel selection, ACS, In-band blocking, out-band blocking, IMD and other receiver specification shall follow the 3GPP guidelines in [3] ^[5] , the reference sensitivity is allowed to degrade at most 6dB under all kinds of interference signal and corresponding level.	Channel selection related requirement	High
Other specifications	Except for all the RF specifications listed above, other RF specifications shall follow the requirement in [3].	Additional standard to comply	High
Downlink modulation Mode	QPSK, 16-QAM, 64-QAM, 256-QAM	DL Modulation schemes	High
Uplink modulation mode	$\pi/2$ -BPSK, QPSK, 16-QAM, 64-QAM	UL Modulation schemes	High
	256QAM		medium
Conducted Power	4T4R: up to 10W per port	RF power	High
	16T16R: TBD		Low

1 4.4.2 O-RU_x Interfaces

2 The interface related requirements of the O-RU_x are listed in Table 4-15.

Table 4-15: O-RU_x Interface Requirements

Parameter	Requirement	Description	Priority
Interface Number	The O-RU _x shall have at least one fronthaul interface based on the split option supported	Number of fronthaul links	High

4.4.3 O-RU_x Environmental and EMC

The environmental and EMC requirements of the O-RU_x are listed in Table 4-16.

Table 4-16: O-RU_x Environmental and EMC Requirements

Parameter	Requirement	Description	Priority
Mounting method	Rooftop, side of building/wall or pole	Mounting requirement	High
Grounding	The O-RU _x shall support Joint Grounding Method and shall work normally when the grounding resistor is less than 10Ω.	Grounding requirement	High
EMC	The O-RU _x shall comply with the requirements of 3GPP TS 38.113 [7]	Electromagnetic Compatibility requirement	High

4.4.4 O-RU_x Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-RU_x are listed in Table 4-17.

1

Table 4-17: O-RU_x Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Weight	4T4R: The gross weight of the O-RU _x shall be <13Kg (best estimate)	Weight requirement	High
	16T16R: TBD		Low
Dimension/Volume	4T4R: The dimension of the O-RU _x shall be smaller than 345x250x130 (mm) and the volume of the O-RU _x shall be <12L	Dimensional Measurements	High
	16T16R: TBD		Low
Stability	The MTBF shall be >200000 hours	Mean Time Between Failures	High
Power Consumption	4T4R: At full load, the power consumption shall not exceed 218W	Power requirement	High
	16T16R: At full load, the power consumption shall not exceed TBD		Low
Power supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300v, frequency range of 45Hz ~ 65Hz.	Power Support Requirement	High
Level of protection	The protection level of the O-RU _x shall be equivalent to the IP65 standard.	Protection level	High
Temperature and Humidity	The O-RU _x shall operate at the temperature range of -40°C to +55°C with both cold start and hot start options. Humidity : 5%~95%	Ambient temperature and moisture requirement	High
Atmospheric pressure	The O-RU _x shall operate normally under atmospheric pressure between 70 to 106Kpa.	Operational atmospheric pressure requirement	High
Cooling mode	Passive cooling	System cooling requirement	High

2

4.5 O-RU_x Split Option Specific Requirements

Besides the common requirements which shall apply to all the radio unit types. The following sections list all the specific requirements that only apply to the designated split option.

4.5.1 O-RU₇₋₂ Specific Requirements

The O-RU₇₋₂ must have one RJ45 or SFP 10G Ethernet interface used as fronthaul interface. The O-RU₇₋₂ must support the lower physical layer functions and comply with [4].

4.5.2 O-RU₆ Specific Requirements

This section intentionally left blank.

4.5.3 O-RU₈ Specific Requirements

This section intentionally left blank.

4.6 FHGW_x – Common Requirements

This section intentionally left blank.

4.7 FHGW_x – Split Option Specific Requirements

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4.8 Integrated gNB-DU Requirements

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EXCEPT FOR BREACH OF CONFIDENTIALITY, ADOPTER’S BREACH OF SECTION 3, AND ADOPTER’S INDEMNIFICATION OBLIGATIONS, IN NO EVENT SHALL ANY PARTY BE LIABLE TO ANY OTHER PARTY OR THIRD PARTY FOR ANY INDIRECT, SPECIAL, INCIDENTAL, PUNITIVE OR CONSEQUENTIAL DAMAGES RESULTING FROM ITS PERFORMANCE OR NON-PERFORMANCE UNDER THIS AGREEMENT, IN EACH CASE WHETHER UNDER CONTRACT, TORT, WARRANTY, OR OTHERWISE, AND WHETHER OR NOT SUCH PARTY HAD ADVANCE NOTICE OF THE POSSIBILITY OF SUCH DAMAGES. O-RAN SPECIFICATIONS ARE PROVIDED “AS IS” WITH NO WARRANTIES OR CONDITIONS WHATSOEVER, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. THE O-RAN ALLIANCE AND THE MEMBERS, CONTRIBUTORS OR ACADEMIC CONTRIBUTORS EXPRESSLY DISCLAIM ANY WARRANTY OR CONDITION OF MERCHANTABILITY, SECURITY, SATISFACTORY QUALITY, NONINFRINGEMENT, FITNESS FOR ANY PARTICULAR PURPOSE, ERROR-FREE OPERATION, OR ANY WARRANTY OR CONDITION FOR O-RAN SPECIFICATIONS.

Section 8: ASSIGNMENT

Adopter may not assign the Agreement or any of its rights or obligations under this Agreement or make any grants or other sublicenses to this Agreement, except as expressly authorized hereunder, without having first received the prior, written consent of the O-RAN ALLIANCE, which consent may be withheld in O-RAN ALLIANCE’s sole discretion. O-RAN ALLIANCE may freely assign this Agreement.

Section 9: THIRD-PARTY BENEFICIARY RIGHTS

Adopter acknowledges and agrees that Members, Contributors and Academic Contributors (including future Members, Contributors and Academic Contributors) are entitled to rights as a third-party beneficiary under this Agreement, including as licensees under Section 3.

Section 10: BINDING ON AFFILIATES

Execution of this Agreement by Adopter in its capacity as a legal entity or association constitutes that legal entity's or association's agreement that its Affiliates are likewise bound to the obligations that are applicable to Adopter hereunder and are also entitled to the benefits of the rights of Adopter hereunder.

Section 11: GENERAL

This Agreement is governed by the laws of Germany without regard to its conflict or choice of law provisions.

This Agreement constitutes the entire agreement between the parties as to its express subject matter and expressly supersedes and replaces any prior or contemporaneous agreements between the parties, whether written or oral, relating to the subject matter of this Agreement.

Adopter, on behalf of itself and its Affiliates, agrees to comply at all times with all applicable laws, rules and regulations with respect to its and its Affiliates' performance under this Agreement, including without limitation, export control and antitrust laws. Without limiting the generality of the foregoing, Adopter acknowledges that this Agreement prohibits any communication that would violate the antitrust laws.

By execution hereof, no form of any partnership, joint venture or other special relationship is created between Adopter, or O-RAN ALLIANCE or its Members, Contributors or Academic Contributors. Except as expressly set forth in this Agreement, no party is authorized to make any commitment on behalf of Adopter, or O-RAN ALLIANCE or its Members, Contributors or Academic Contributors.

In the event that any provision of this Agreement conflicts with governing law or if any provision is held to be null, void or otherwise ineffective or invalid by a court of competent jurisdiction, (i) such provisions will be deemed stricken from the contract, and (ii) the remaining terms, provisions, covenants and restrictions of this Agreement will remain in full force and effect.
